SEC Analysis Report

De Novo Designed Heterodimeric Protein Ring Assemblies

Date: April 2026
Analyst: BioClaw
Project: Chain A (~53 kDa) + Chain B (~27 kDa) co-purified constructs

Primary cohort	6
Context runs	0
Detection	UV 280 nm
Column	Superdex 200 Increase 10/300 GL (V0 ~ 8.0 mL)
Report generated	2026-04-16
Analysis	Automated (Python / SciPy / Typst)

Executive Summary

- Primary analysis focuses on 6 construct(s) from .
- Best-behaved controls are Monomer-Only-09, Dimer-Variant-03.
- High-confidence HMW lead(s): Ring-Design-01, Ring-Design-11.

1. Background

This report presents a size-exclusion chromatography (SEC) analysis of 6 de novo designed protein constructs intended to form specific oligomeric assemblies, including dimers and ring-like higher-order structures. Each sample consists of co-purified Chain A (481 amino acids, 53 kDa) and Chain B (243 amino acids, 27 kDa), forming a heterodimer of approximately 80 kDa.

SEC provides a rapid, solution-phase readout of the oligomeric state distribution. Constructs that assemble into rings are expected to elute early (large apparent size), while failed or partially assembled constructs will elute as dimers, monomers, or aggregates. This analysis covers 6 constructs with elution zones classified relative to a void volume of 8.0 mL.

Note
No absolute MW calibration was applied. All size assignments are relative, based on elution position relative to the column void volume. The terms 'HMW', 'mid-size', and 'small/monomer' refer to relative elution zones only.

2. Methods

2.1 Instrumentation and Columns

Purified protein samples were injected onto an SEC column (inferred to be a Superdex 200 Increase 10/300 GL or equivalent, 24 mL bed volume), AKTA system, UV detection at 280 nm. Elution volumes reported relative to sample application anchor (0 mL = injection point). Void volume estimated at 8.0 mL.

2.2 Data Processing

- Savitzky-Golay smoothing (window = 15 points / 0.75 mL, polynomial order 3) to reduce instrument noise.
- Automated peak detection using SciPy `find\_peaks` with adaptive height threshold (3% of maximum signal or 1.0 mAU, whichever is greater), minimum prominence of 1.5 mAU, and minimum inter-peak distance of 0.5 mL.
- Peak area integration by trapezoidal rule between 10%-height boundaries.
- Species classification based on elution volume relative to void volume.

2.3 Elution Zone Definitions

ZONE	VOLUME RANGE (ML)	INTERPRETATION
Void / Aggregate	0.0 -- 8.0	Aggregates or void-excluded species
Very Large (HMW)	8.0 -- 10.5	Ring candidates (large oligomers)
Large Oligomer	10.5 -- 12.5	Intermediate oligomers
Mid-size	12.5 -- 14.5	Dimer / tetramer range
Small / Monomer	14.5 -- 20.0	Monomer-like species

Elution zone definitions.

3. Results

3.1 Overview

6	0	3	10.0
PRIMARY COHORT	CONTEXT RUNS	HMW FOLLOW-UP	BEST QUALITY

3.2 Construct Summary Table

CONSTRUCT	DOM. PEAK (ML)	DOM. ZONE	DOM. FRAC.	VOID FRAC.	CLASSIFICATION
Monomer-Only-09	15.5	Small / Monomer	100%	0%	Best control
Ring-Design-01	10.0	Very Large (HMW)	100%	0%	HMW lead
Ring-Design-11	9.8	Very Large (HMW)	85%	0%	HMW lead
Dimer-Variant-03	13.5	Mid-size	81%	0%	Best control
Mixed-Assembly-07	11.0	Large Oligomer	31%	0%	HMW follow-up
Aggregator-05	8.0	Void / Aggregate	50%	50%	Aggregation

Primary-cohort constructs ranked for follow-up decisions.

3.3 SEC Overlay Comparison

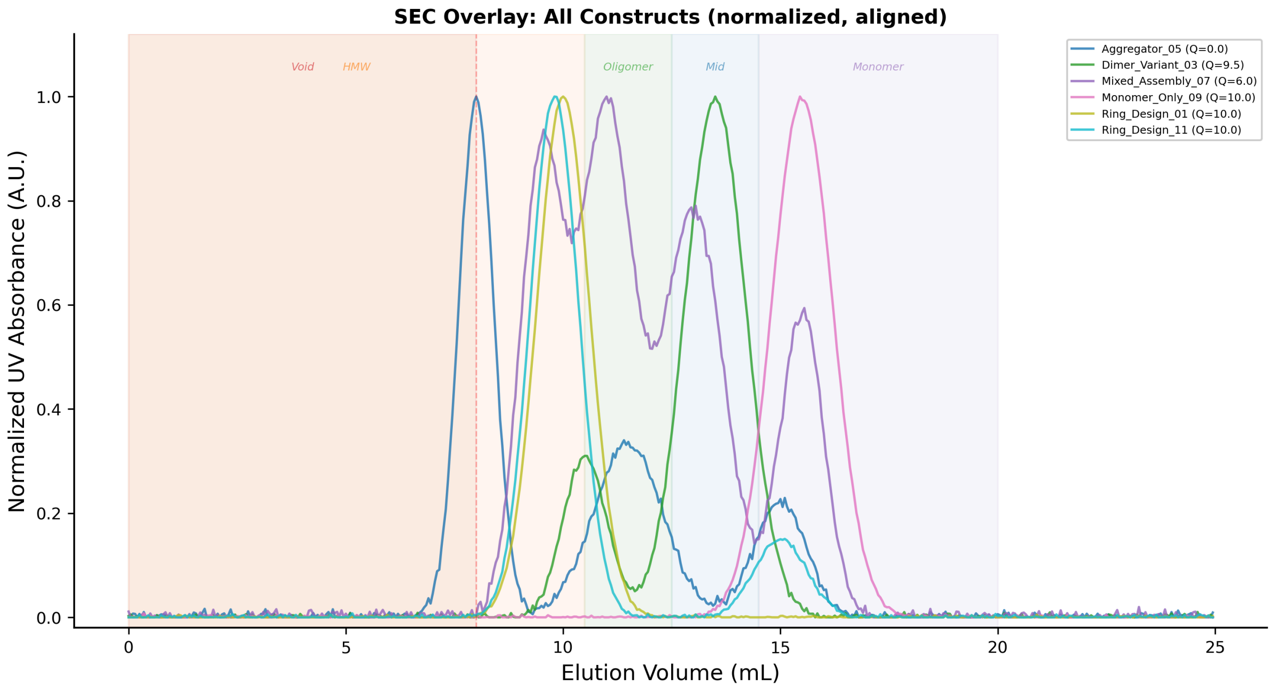

Figure 1: Primary cohort overlay (6 constructs), normalized and aligned to the injection point.

3.4 Zone Fraction Analysis

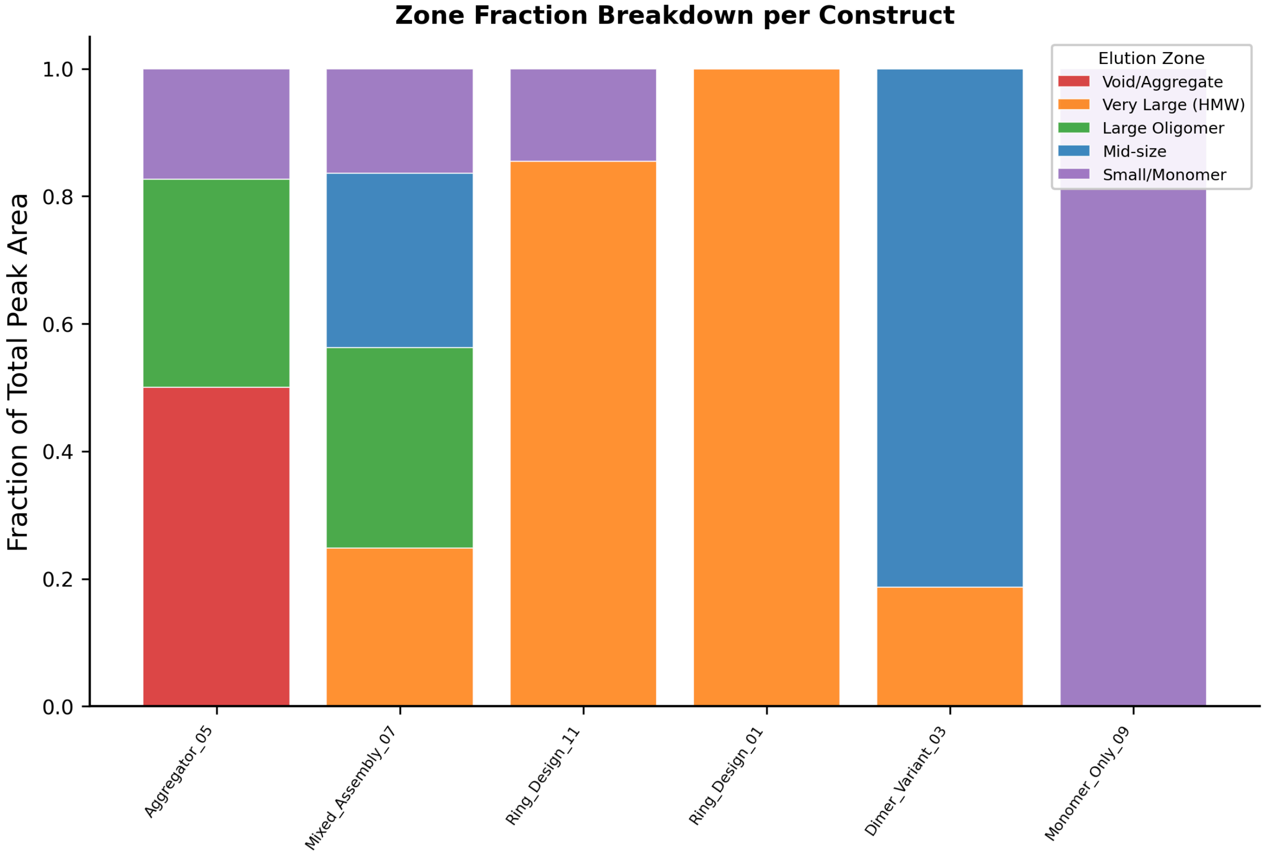

Figure 2: Primary-cohort zone fractions, sorted by dominant peak position.

3.5 Quality Ranking

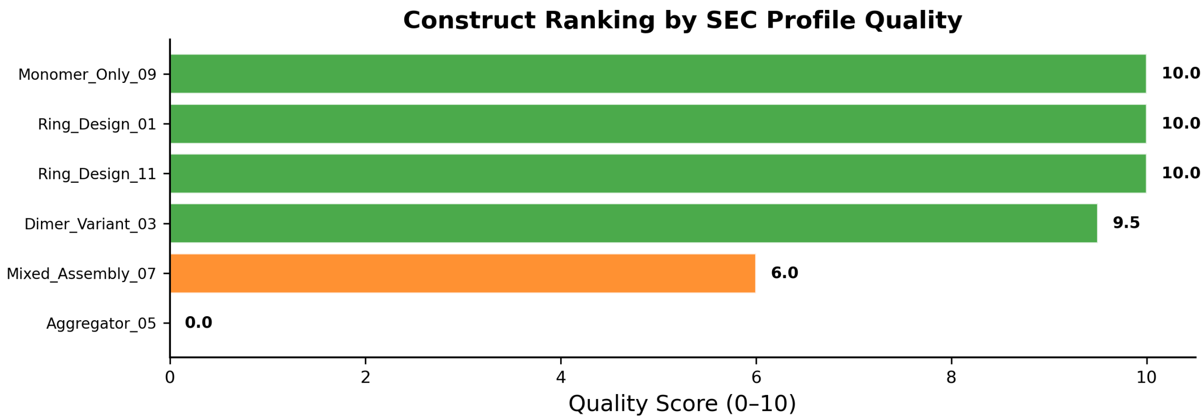

Figure 3: Primary-cohort constructs ranked by SEC profile quality score (0–10).

3.6 Representative Chromatograms

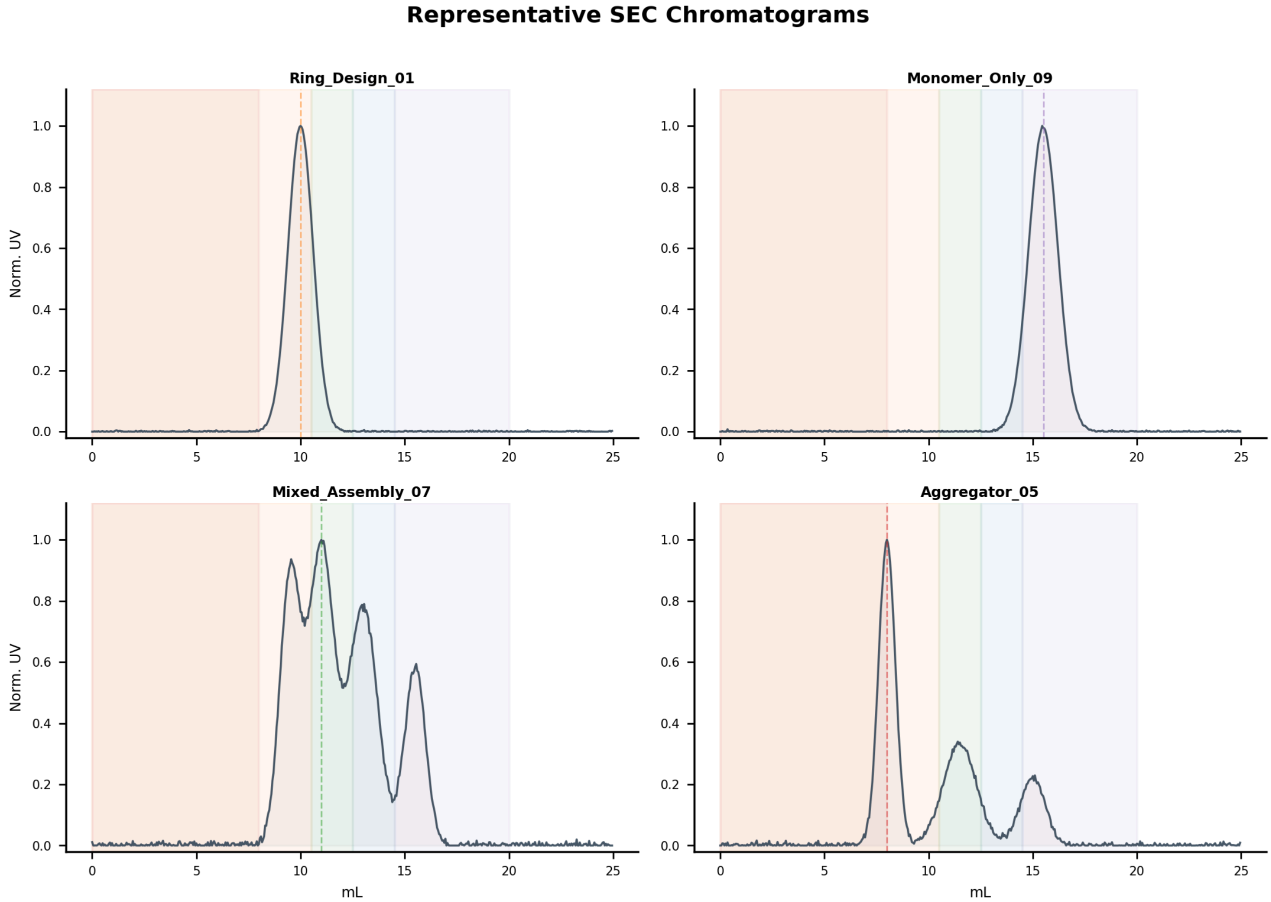

Figure 4: Representative primary-cohort chromatograms spanning clean controls, HMW-leading traces, heterogeneous profiles, and late-eluting failures.

3.7 Selected Construct Highlights

This compact report highlights 4 representative constructs out of 6 total. Complete per-construct chromatograms remain available in supporting outputs.

Ring-Design-01 – Excellent *

Dominant peak at 10.0 mL (Very Large (HMW) zone), 100% of total area, FWHM = 1.42 mL. Highly monodisperse profile. Elution in the HMW zone is consistent with higher-order oligomeric assembly.

Recommendation: Priority: cryo-EM / native MS

Monomer-Only-09 – Excellent *

Dominant peak at 15.5 mL (Small / Monomer zone), 100% of total area, FWHM = 1.65 mL. Highly monodisperse profile.

Recommendation: Monomer control; SEC-MALS for MW

Mixed-Assembly-07 – Good

Dominant peak at 11.0 mL (Large Oligomer zone), 31% of total area, FWHM = 4.70 mL. Secondary peak(s) at 9.6 mL (25%), 13.0 mL (27%), 15.5 mL (16%). Polydisperse — multiple species present. Broad peak(s) suggest conformational heterogeneity or equilibrium between oligomeric states. Elution in the HMW zone is consistent with higher-order oligomeric assembly.

Recommendation: Buffer optimisation

Aggregator-05 – Poor

Dominant peak at 8.0 mL (Void / Aggregate zone), 50% of total area, FWHM = 0.96 mL. Secondary peak(s) at 11.4 mL (33%), 15.0 mL (17%). Significant void-volume signal (50%) indicates non-specific aggregation. Polydisperse — multiple species present.

Recommendation: Redesign interface

4. Discussion

4.1 Ring Assembly Candidates

The clearest ring assembly candidates are Ring_Design_01, Ring_Design_11, Mixed_Assembly_07. These show dominant peaks in the HMW zone (8.0–12.5 mL), consistent with higher-order oligomeric assemblies substantially larger than the expected 80 kDa heterodimer.

Several candidates (Mixed_Assembly_07) show broad FWHM in the HMW peak, which may reflect a mixture of oligomeric states (e.g., rings of different stoichiometries) rather than a single discrete ring. This is common for de novo designed rings where the assembly equilibrium may favour a distribution of ring sizes.

4.2 Monodisperse Assemblies

The most monodisperse samples in the dataset are Monomer_Only_09, Ring_Design_01, Ring_Design_11. These constructs show sharp, symmetric peaks with minimal secondary species, making them excellent candidates for structural characterisation (SEC-MALS, cryo-EM, or crystallography).

4.3 Heterogeneous HMW Assemblies

Several constructs (Mixed_Assembly_07) show HMW-dominant profiles but with broad, multi-modal distributions. These may represent: (1) a mixture of ring sizes in equilibrium, (2) partially assembled intermediates, or (3) non-specific oligomerisation. Distinguishing between these possibilities requires orthogonal techniques (native PAGE, DLS, or cryo-EM). The low void fractions argue against simple aggregation.

4.4 Aggregation Concerns

Aggregator_05: 50% void-volume signal. Buffer optimisation or interface redesign recommended.

4.5 Methodological Considerations

- No absolute MW calibration: all size assignments are relative to void volume position.
- Peak metrics (FWHM, area fraction) depend on smoothing parameters and baseline correction.
- SEC-MALS would be required for definitive molecular weight determination.
- Normalisation: all traces are normalised to their own maximum UV signal; absolute signal levels (protein concentrations) are not compared between constructs.

5. Conclusions and Recommendations

5.1 Ranked Construct Summary

RANK	CONSTRUCT	CLASSIFICATION	RECOMMENDATION
1	Monomer-Only-09	Small / Monomer	Monomer control; SEC-MALS for MW
2	Ring-Design-01	Very Large (HMW)	Priority: cryo-EM / native MS
3	Ring-Design-11	Very Large (HMW)	Priority: cryo-EM / native MS
4	Dimer-Variant-03	Mid-size	Use as positive control
5	Mixed-Assembly-07	Large Oligomer	Buffer optimisation
6	Aggregator-05	Void / Aggregate	Redesign interface

Final construct ranking with recommendations.

5.2 Recommended Next Experiments

Priority Actions

For ring candidates (Ring_Design_01, Ring_Design_11):

- Native PAGE – rapid check for discrete oligomeric bands vs. smear
- DLS (Dynamic Light Scattering) – measure hydrodynamic radius and polydispersity index (PDI < 0.2 supports a discrete ring)
- Cryo-EM – definitive structural characterisation; collect and concentrate the HMW SEC fraction
- Native mass spectrometry – determine exact stoichiometry of the ring

Secondary Actions

For monodisperse constructs (Monomer_Only_09, Dimer_Variant_03):

- SEC-MALS – determine absolute MW of the dominant peak
- Cryo-EM or X-ray crystallography – high-resolution structure
- Analytical ultracentrifugation (AUC) – confirm oligomeric state in solution

Optimisation

For heterogeneous HMW constructs (Mixed_Assembly_07):

- Buffer screen – vary salt concentration, pH, and additives (glycerol, detergent)
- Temperature screen – some ring assemblies are temperature-sensitive
- SEC-MALS on the HMW fraction – determine whether the HMW material is a discrete species

Disclaimer

Report generated by BioClaw. All data sourced from user-uploaded AKTA CSV files. No absolute molecular weight calibration was applied.